

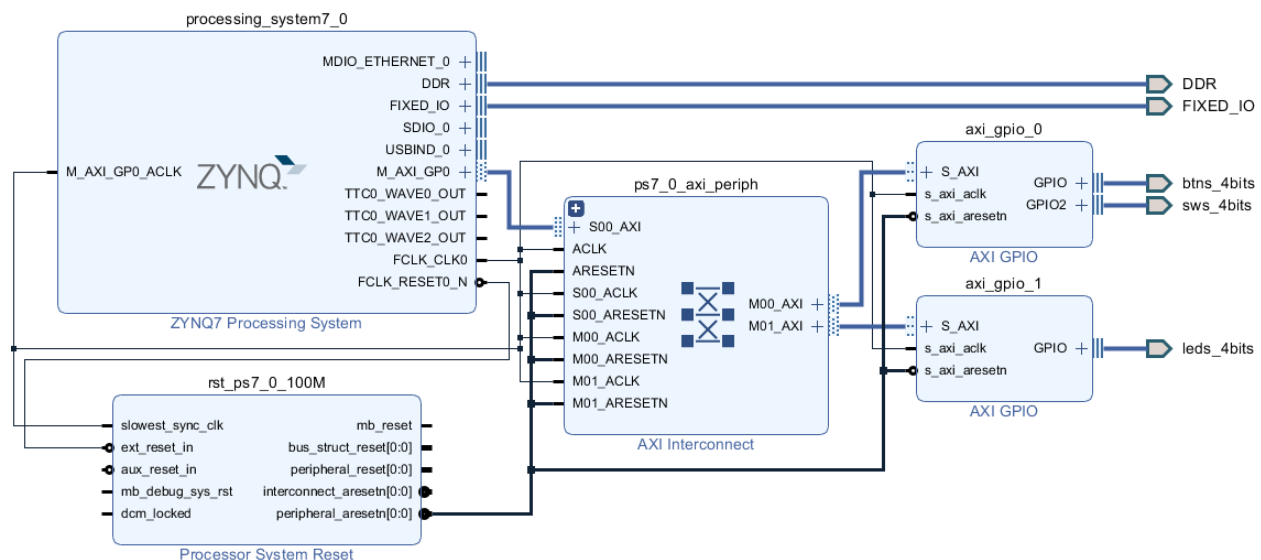
ECE3623 Embedded System Design Laboratory



Task Management in FreeRTOS on Zybo

In this Lab, you will investigate task management in the FreeRTOS environment on the *fchannel* for the Zybo push buttons and axi_GPIO_1 as a single channel for the Zybo LEDs as shown below.

In this Lab, you are modifying that Vivado hardware design with axi_gpio_0 as a dual channel for the Zybo push buttons on channel 0 and slide switches on channel 1, as shown below. This Vivado hardware design will be used in this and several Labs to follow.



The BTN's and SW's GPIO are at *XPAR_AXI_GPIO_0_DEVICE_ID* but the GPIO has two channels. The GPIO function calls for the BTN's and SW's must specify the correct channel number (0 or 1). The LEDs GPIO are at *XPAR_AXI_GPIO_1_DEVICE_ID*. The GPIO function calls for the LEDs must specify the correct channel number (0).

Launch SDK and search and add the *freertos_hello_world.c* application as described in the *Introduction to FreeRTOS on Zybo* posted on Canvas. You are to modify *freertos_hello_world.c* to be the new *main()* application (using another name) that performs the following FreeRTOS task management operations with the Vivado hardware design. Engineering projects are often stated as procedures that must be rendered as process control software on the appropriate embedded hardware.

1. In Main, create (*xTaskCreate()*) a FreeRTOS task **TaskLED** at priority 1 that is a 4-bit counter for illuminating the four LEDs.
2. In Main, create (*xTaskCreate()*) a FreeRTOS task **TaskSWS** that reads SW0 at a priority of 1. Within TaskSWS, the state of SW0 toggles the counter in TaskLED between being an up-counter and a down-counter. Hint: SW0 should toggle a global variable used by TaskLED.
3. In Main, create (*xTaskCreate()*) a FreeRTOS task **TaskBTN** that reads BTN0, BTN1, BTN2 and BTN3 at priority 1.

Within TaskBTN:

- a) If BTN0 is depressed, then *TaskLED* is suspended (*vTaskSuspend()*), and if BTN0 is released, then *TaskLED* is resumed (*vTaskResume()*).
- b) If BTN1 is depressed, then *TaskSWS* is suspended (*vTaskSuspend()*), and if BTN1 is released, then *TaskSWS* is resumed (*vTaskResume()*).
- c) If BTN2 and BTN3 are both depressed simultaneously and released, then both *TaskLED* and *TaskSWS* are temporarily suspended (*vTaskSuspend()*) for 5 seconds and then automatically resumed (*vTaskResume()*). Hint: lookup *vTaskDelay()*.

Note: The FreeRTOS the task management functions *vTaskSuspend()* and *vTaskResume()* utilize the “*handles*” of the created tasks.

